# Omar Fadlallah

Machine Learning Engineer | AI Engineer (NLP & RAG Systems)

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Portfolio • LinkedIn • GitHub

## Summary

Machine Learning Engineer specializing in LLMs and Retrieval-Augmented Generation (RAG), building scalable, low-latency AI systems with measurable impact (sub-50ms latency, up to 80% cost reduction). Strong expertise in LLMOps, distributed system design, and production ML infrastructure.

## Education

**Bachelor's Degree in Computer Science (Ongoing) at Faculty of Computers and Information, Mansoura University, Mansoura | October 2022 - Expected June 2026**

**Grade:** 3.7/4.0

## Projects

**Enterprise Multi-Tenant Agentic RAG Infrastructure (Production-Oriented)**
**GitHub**: **https://github.com/mirofadlalla/Atlas-AI-Platform.git**

- Architected a production-grade multi-tenant Agentic RAG system, enabling enterprise-scale document QA with secure JWT authentication and strict tenant isolation.
- Designed a hybrid retrieval & ingestion pipeline (Dense + BM25 + reranking) with Qdrant indexing, improving QA accuracy and reducing irrelevant responses.
- Engineered LLM pipelines with streaming & cost tracking, improving response efficiency and enabling real-time cost and latency monitoring.
- Fine-tuned Qwen2.5-1.5B using LoRA/QLoRA for Arabic instruction-following, improving response quality, Arabic understanding, and domain adaptation.
- Optimized system performance via 3-tier caching (RAM, Redis semantic, DB), reducing latency to <50ms and lowering LLM costs by 60–80%.
- Developed agent-based reasoning system with multi-step execution, integrated with full observability (Prometheus, Grafana) for latency, token usage, and telemetry tracking.

**DeepRAG – MLOps-Ready Retrieval-Augmented Generation System**
**GitHub: https://github.com/mirofadlalla/DeepRAG**

- Designed an end-to-end RAG system with a modular retrieval pipeline (FAISS IVF, BM25, Cross-encoder re-ranking) using BAAI/bge-m3 semantic embeddings.
- Implemented rigorous retrieval evaluation metrics, achieving significant improvements in MRR (+60%), Recall@K, and Precision@K (+80%).
- Integrated MLflow for MLOps workflows, managing experiment tracking, metrics logging, and model artifacts.
- Developed automated evaluation modules for hallucination detection, answer relevance, and strict answer grounding (numeric/unit preservation).
- Established system reliability monitoring with drift detection (cosine similarity) and retrieval stability tracking (Jaccard similarity).

**LLMOps Platform – Production-Grade LLM Prompt Management**
**GitHub**: **https://github.com/mirofadlalla/LLM-Platform-LLMOps-System.git**

- Designed and containerized a FastAPI-based backend (Docker) for managing, testing, and evaluating LLM prompts at scale.
- Built a modular evaluation engine featuring LLM-based scoring, hallucination detection, regression testing, and prompt versioning.
- Engineered robust async workflows using Celery for large-scale experiments, and managed PostgreSQL databases using SQLAlchemy ORM and Alembic.
- Applied backend best practices, including structured logging, API authentication, request tracing, and secure deployments.

## Internships

**AI Engineer — Devoteam (Intern), Remote Belgium |** Dec 2025 – Feb 2026

- Fine-tuned Transformer-based LLMs using PyTorch, improving model accuracy on text classification and summarization tasks by ~15% compared to baseline models.
- Engineered robust text preprocessing pipelines for RAG setups, successfully parsing and cleaning over 50,000+ document chunks to enhance vector search quality.
- Contributed to 2 enterprise AI proof-of-concepts, leveraging advanced prompt engineering to reduce irrelevant responses and streamline natural language workflows.

**Generative AI Trainee — NVIDIA Deep Learning Institute & ITI ,** Cairo | August 2025 - October 2025

- Developed 3 end-to-end GenAI use cases focusing on LLMs and RAG architectures as part of an intensive applied training program.
- Implemented advanced prompt engineering techniques, optimizing model context windows and reducing hallucination rates in generated responses by ~20%.
- Earned 2 official NVIDIA Academy Certifications, validating hands-on expertise in applied Deep Learning, Generative AI, and NLP fundamentals.

## Skills

**LLMs & RAG: Transformers, RAG Architectures, Prompt Engineering, LangChain, LangGraph**
**MLOps & Systems: FastAPI, Docker, MLflow, Redis, Celery**
**Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost**
**Data: Pandas, NumPy, Power BI**
**Databases: PostgreSQL, MySQL, MongoDB, FAISS**
**Programming: Python, SQL, C++, R**

## Languages

**Arabic – Native**
**English - Fluent**